拥挤的苍穹

The Crowded Heavens



两颗卫星相撞，为什么多年后仍要避让？

英文原文

On 10 February 2009, nearly 790 kilometres above Siberia, two satellites **slammed** together at about 42,000 kilometres an hour. Iridium 33 **belonged to** an American satellite-phone **constellation**; Kosmos 2251, a Russian military satellite, had been **derelict** since about 1995. Both **shattered** in the first **accidental collision** between two intact satellites.

Neither craft **swerved**. The Russian **hulk**, lacking **propulsion**, could not. A public collision-warning service had forecast a **narrow miss** of about 584 metres, yet the predicted **encounter** was not among the week's closest; such warnings were **frequent** and **imprecise**. One owner's **discarded relic demolished** another's working **asset**.

The **wreckage**, however, **endangered** everyone else. The **impact flung out** more than 2,000 **catalogued** fragments and **a multitude of** others too small to **pinpoint**. At that height **air drag** is **feeble**, so many would **linger** for decades. In January 2012, the International **Space Station fired** its **thrusters** to **dodge** an Iridium 33 **remnant**.

In June 1978, two NASA scientists **envisaged** a **vicious spiral**, later named the Kessler **syndrome**: each collision would **scatter** fragments, **raising the odds of subsequent** crashes. The 2009 crash **illustrated** that **premise** without **triggering** a **runaway chain reaction**. By the end of 2024, **surveillance** networks tracked about 40,000 objects, only some 11,000 of them **functional**.

英文原文（续）

Such a cascade would **unfold** over decades, though, and is not **inevitable**; the risk can be **mitigated**. Operators now **routinely manoeuvre** satellites **out of harm's way**. In September 2022, America's Federal Communications Commission, which licenses satellites, **mandated disposal** of low-orbit craft within five years of **retirement instead of** 25. Years after the crash, spacecraft whose owners **had no part in** it kept swerving around its **splinters**.

中文译文

2009年2月10日，在西伯利亚上空近790公里处，两颗卫星猛撞在一起，时速约42,000公里。铱星33号属于美国的一个卫星电话星座；宇宙2251号是一颗俄罗斯军用卫星，约自1995年起便已废弃。两星双双粉碎，这是首次有意外相撞发生在两颗完整卫星之间。

两个飞行器都没有紧急转向避让。那个俄罗斯的（卫星）残骸，由于缺乏推进力，无法转向。一项公开的碰撞预警服务曾预报，两星将惊险擦过，相距约584米，但预测中的这次交会并未排进当周距离最近的几次之列；这类预警既频繁又不精确。一方弃置的遗物，意外毁坏了另一方仍在工作的资产。

这些残骸却危及了所有其他人。撞击抛出了2,000多块已编目的碎片，另有大量碎片太小，无法精确定位。在那个高度，空气阻力微弱，因此许多碎片会在轨道上滞留数十年。2012年1月，国际空间站点火了它的推进器，以躲开/避开一块铱星33号的残片。

1978年6月，两位美国国家航空航天局（NASA）的科学家设想了一种恶性循环，后来被称为凯斯勒综合征：每次相撞都会散布碎片，提高了后续相撞的几率。2009年的这次相撞例证了这一前提，却并未引发失控的连锁反应。截至2024年底，太空监视网络跟踪了约40,000个物体，其中只有约11,000个能正常运行。

不过，这样的连锁效应会逐步展开，历时几十年，而且并非不可避免，风险也可以加以缓解。运营方如今会例行操纵卫星避开危险。2022年9月，负责为卫星发放许可的美国联邦通信委员会强制要求，低轨航天器须在退役后五年内处置，而不是25年。相撞多年之后，那些所有者与此事毫无瓜葛的航天器，仍不得不一再转向，绕开那次相撞留下的碎片。

词汇与表达

slammed v.
猛撞

belonged to v.
属于

constellation n.
星座

derelict adj.
废弃
had been derelict since about 1995

shattered v.
粉碎

accidental adj.
意外

collision n.
相撞
the first accidental collision

swerved v.
紧急转向避让

hulk n.
残骸

propulsion n.
推进力
lacking propulsion

narrow miss adj.
惊险擦过
a narrow miss of about 584 metres

encounter n.
交会

frequent adj.
频繁

imprecise adj.
不精确
frequent and imprecise

discarded adj.
弃置的
One owner's discarded relic

relic n.
遗物

demolished v.
意外毁坏了

asset n.
资产
another's working asset

wreckage n.
残骸
The wreckage, however

endangered v.
危及了

impact n.
撞击

a multitude of phr.
大量

flung out v.
抛出了

catalogued adj.
已编目的
more than 2,000 catalogued fragments

pinpoint v.
精确定位
too small to pinpoint

air drag n.
空气阻力
air drag is feeble

feeble adj.
微弱

linger v.
滞留
many would linger for decades

词汇与表达（续）

Space Station n.
空间站

fired v.
点火了

thrusters n.
推进器

dodge v.
躲开/避开
to dodge an Iridium 33 remnant

remnant n.
残片

envisaged v.
设想了
scientists envisaged a vicious spiral

vicious spiral n.
恶性循环

syndrome n.
综合征

scatter v.
散布

raising the odds of v.
提高了……的几率

subsequent adj.
后续
the odds of subsequent crashes

illustrated v.
例证了

premise n.
前提
illustrated that premise

triggering v.
引发
without triggering

runaway adj.
失控的

chain reaction n.
连锁反应
a runaway chain reaction

surveillance n.
监视
surveillance networks tracked

functional adj.
正常运行
11,000 of them functional

unfold v.
展开

inevitable adj.
不可避免
is not inevitable

mitigated v.
加以缓解

routinely adv.
例行

out of harm's way phr.
避开危险

manoeuvre v.
操纵

mandated v.
强制要求

disposal n.
处置
mandated disposal of low-orbit craft

retirement n.
退役

instead of prep.
而不是

had no part in v.
与……毫无瓜葛

splinters n.
碎片

文章简述

- 相撞：2009年，两颗完整卫星首次意外相撞：失效的宇宙2251号撞上仍工作的铱星33号。
- 预警：公开预警频繁且不准，预计两星会隔约584米擦过，并非当周最近交会；失效卫星也无法避让。
- 碎片：碰撞留下两千多块被登记追踪的碎片；截至2024年底，全球跟踪约4万个在轨物体，只有约1.1万个正常运行。
- 风险：高空空气阻力小，碎片能留上数十年，还可能撞出更多碎片；此次没有触发失控连锁反应，恶性循环并非必然。
- 应对：运营方加强卫星避让；美国联邦通信委员会要求其监管的低轨卫星，退役后5年内离轨处置，缩短了原来的25年期限。